

**Gary Miller**

United States | Remote /Open to Relocation  
[LinkedIn](#) | [GitHub](#) | [GM Cubed Designs](#)

## Professional Summary

Lead Applied AI Engineer with over 7 years of full-stack software engineering experience and a deep specialization in designing autonomous, multi-agent AI systems. Demonstrated expertise in local LLM orchestration, retrieval-augmented generation (RAG) architectures, and deterministic data pipelines. Proven track record in adversarial testing (Red Teaming) of frontier models, utilizing iterative RLHF to expose and patch critical safety vulnerabilities. Adept at engineering controlled, simulated environments to evaluate autonomous behavior and architecting localized systems that strictly enforce enterprise data privacy. United States Navy Veteran equipped with a strong foundation in high-stakes operational discipline, strategic problem-solving, and continuous process improvement.

## Core Competencies & Technical Skills

- **AI & Machine Learning:** Local LLM Orchestration (Ollama, Hugging Face), Open-Source Models (Gemma, Llama, Qwen), RAG Architectures, Vector Databases (Qdrant), Multi-Agent Systems, Generative AI Evaluation, Deterministic JSON Pipelines, AI Red Teaming, Adversarial Testing, RLHF, Safety Alignment, Vulnerability Discovery.
- **Quantitative Finance:** Financial Modeling, Price Prediction, Algorithmic Trading APIs (Kraken), Quantitative Backtesting.
- **Programming Languages:** Python, TypeScript, JavaScript, Ruby, SQL, HTML, CSS, C, Go.
- **Frameworks & Technologies:** React, Node.js, Ruby on Rails, Git/GitHub, Docker, AWS.
- **Methodologies:** Agile Software Development, DevOps, CI/CD, Unit Testing, Big O Notation & Asymptotic Analysis.

## Professional Experience

**G M Cubed Designs** | Lead AI & Quantitative Software Engineer / Founder | Apr 2019 – Present

- Design and orchestrate localized AI ecosystems utilizing Ollama and Qdrant to prioritize **enterprise-grade data privacy and secure containment** within highly complex and sensitive information environments.
- Engineer **high-fidelity controlled environments for assessing autonomous multi-agent systems**, integrating real-time market data and quantitative backtesting to evaluate strategic decision-making under risk.
- Delivered rigorous technical execution resulting in a 56.27% year-over-year increase in gross revenue through independent AI research, architectural innovation, and strategic deployment.
- Conducted systematic adversarial testing (Red Teaming) across frontier LLMs to expose critical safety vulnerabilities and successfully bypass alignment guardrails via deep prompt engineering.

- Authored technical vulnerability reports and engineered iterative Human Reinforcement Learning (RLHF) to patch exploits to directly inform safety alignment retraining.
- Direct comprehensive full-stack development lifecycles using Python and TypeScript to architect scalable, resilient, and high-performance software solutions.

#### US Navy | Aviation Structural Mechanic | Feb 2007 – Jun 2010

- Performed vital, high-precision maintenance and inspections on jet aircraft, managing critical hydraulic, structural, and landing gear systems.
- Executed operational protocols with strict adherence to safety and regulatory standards, handling hazardous materials and maintaining detailed inspection records.

## Select Technical Projects

- **Holdem-Trainer:** Architected a high-fidelity validation framework consisting of 700+ custom unit tests designed to proactively identify regressions and secure system reliability for autonomous strategic logic.
- **Sidekick LLM:** Engineering a transition from a localized chat client into a fully autonomous, multi-agent software engineering agency capable of collaborative problem-solving and code generation.
- **Investment Matrix:** Developed an institutional-grade financial modeling and price prediction application designed to analyze and backtest the crypto market using advanced quantitative metrics.
- **SYNAPSE:** Architected a highly decoupled, multi-agent orchestration framework driven by an asynchronous data pipeline, demonstrating advanced ML systems engineering and scalable infrastructure design for autonomous AI workloads.

## Technical Writing & Publications

#### Author / Contributor | [Medium](#)

- Authored "Quantization in AI: Mastering Precision for Enterprise Efficiency," detailing advanced optimization architectures for deploying massive LLMs at scale, including INT8/INT4 precision scaling, Post-Training Quantization (PTQ), and Quantization-Aware Training (QAT) for edge AI environments.
- Authored comprehensive technical guides on full-stack development, including deep dives into Big O Notation and Asymptotic Analysis, Metaprogramming in Ruby, and API Integration.

## Education

- **Harvard CS50** | Computer Science Online Course (2023)
- **App Academy** | Bootcamp in Full Stack Software Engineering (2019)
- **Austin Community College** | Associate Degree, Honors Program (2013 – 2015) -Econ

## Certifications

- **Lean Six Sigma Green Belt (SSGB)** | American Society of Quality (ASQ)
- **Introduction to Cloud 101** | Amazon Web Services (AWS)